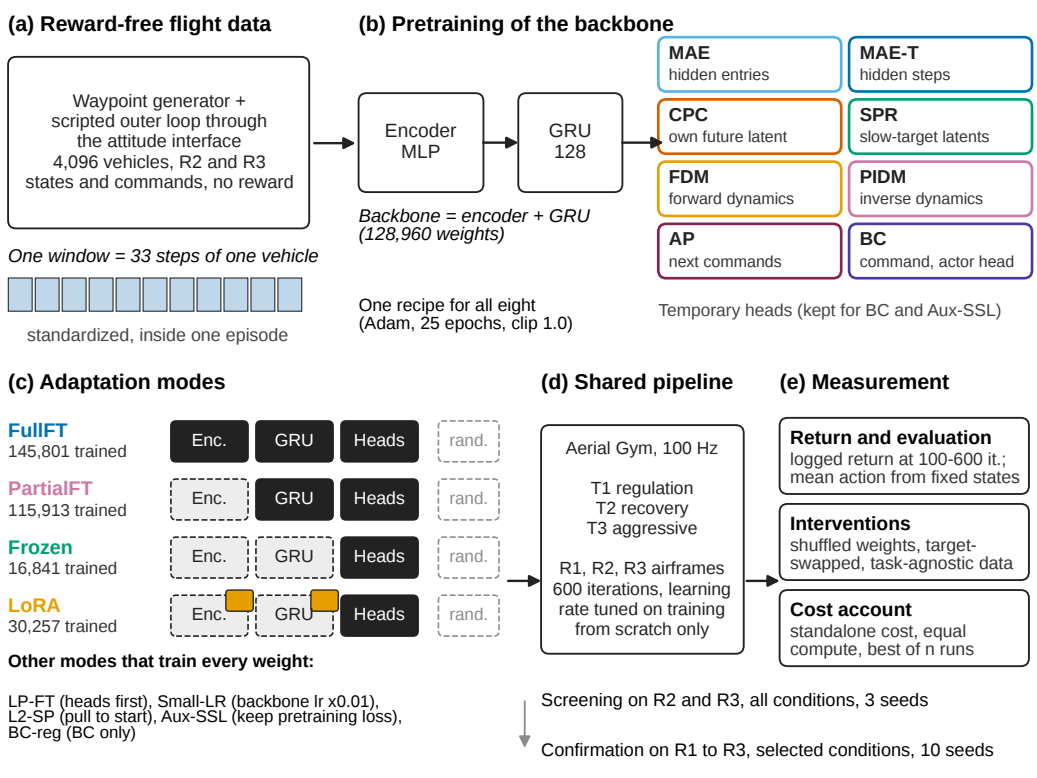
(a) Reward-free flight data
Waypoint generator + scripted outer loop through the attitude interface
4,096 vehicles, R2 and R3
states and commands, no reward
One window = 33 steps of one vehicle
standardized, inside one episode
(b) Pretraining of the backbone
Encoder MLP
GRU 128
Backbone = encoder + GRU (128,960 weights)
One recipe for all eight (Adam, 25 epochs, clip 1.0)
MAE
hidden entries
MAE-T
hidden steps
CPC
own future latent
SPR
slow-target latents
FDM
forward dynamics
PIDM
inverse dynamics
AP
next commands
BC
command, actor head
Temporary heads (kept for BC and Aux-SSL)
(c) Adaptation modes
FullFT
145,801 trained
PartialFT
115,913 trained
Frozen
16,841 trained
LoRA
30,257 trained
Enc.
GRU
Heads
rand.
Other modes that train every weight:
LP-FT (heads first), Small-LR (backbone lr x0.01), L2-SP (pull to start), Aux-SSL (keep pretraining loss), BC-reg (BC only)
(d) Shared pipeline
Aerial Gym, 100 Hz
T1 regulation
T2 recovery
T3 aggressive
R1, R2, R3 airframes
600 iterations, learning rate tuned on training from scratch only
Screening on R2 and R3, all conditions, 3 seeds
Confirmation on R1 to R3, selected conditions, 10 seeds
(e) Measurement
Return and evaluation
logged return at 100-600 it.; mean action from fixed states
Interventions
shuffled weights, target-swapped, task-agnostic data
Cost account
standalone cost, equal compute, best of n runs